

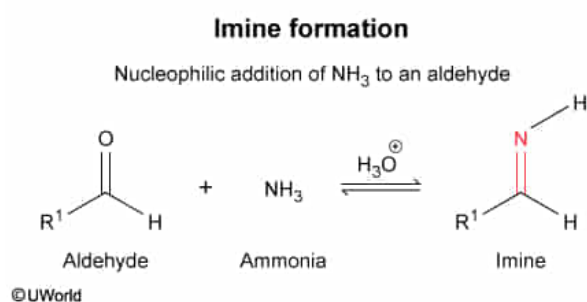
What types of functional groups react to form the imine intermediate in the Strecker synthesis of amino acids?

- ✔ ☒ A. An aldehyde and ammonia [45%]
- ☐ B. An aldehyde and a secondary amine [38%]
- ☐ C. A ketone and a tertiary amine [14%]
- ☐ D. A ketone and a quaternary amine [1%]

Correct

45% answered correctly

Explanation:



Ketones and aldehydes react with ammonia (NH_3) or primary (1°) amines to form **imines**, which are ketone and aldehyde analogues that contain a carbon-nitrogen double bond. The reaction begins with an acid-catalyzed **addition of the amine** to the carbonyl, which involves protonation of the carbonyl, nucleophilic attack of the carbonyl by the amine, and deprotonation of the amine. This forms an intermediate known as an α -aminoalcohol. The second part of the reaction is an **acid-catalyzed dehydration** involving protonation of the $-\text{OH}$ group, loss of H_2O , and deprotonation.

The **Strecker synthesis** is used to make α -amino acids from aldehydes using NH_3 and potassium cyanide (KCN). The first step of the reaction proceeds with protonation of the carbonyl oxygen by H_3O^+ , followed by nucleophilic attack of the carbonyl carbon by NH_3 , resulting in dehydration and imine formation. Therefore, an **aldehyde and NH_3** are used to form the imine intermediate in the Strecker synthesis.

(Choice B) An aldehyde and a secondary amine react to form an **enamine** via an acid-catalyzed addition of the amine followed by an **acid-catalyzed dehydration**. This reaction does *not* form an imine.

(Choices C and D) Ketones can be used in the formation of imines. However, **tertiary and quaternary amines** are not suitable for imine synthesis because tertiary amines do not have an available hydrogen for deprotonation and quaternary amines

cannot act as nucleophiles and add the ketone (or aldehyde) carbonyl.

Educational objective:

An imine is an analogue of ketones and aldehydes that contains a carbon-nitrogen double bond. Imines are formed from a ketone or aldehyde and NH_3 or a primary amine via an acid-catalyzed addition of the amine followed by an acid-catalyzed dehydration.

Time spent:0 sec

Subject:

Organic Chemistry

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System:

Functional Groups and Their Reactions